

FORT ASSINIBOINE AGCM, AB, Canada

WMO: 714830

Lat: 54.410N

Lon: 114.769W

Elev: 2218

StdP: 13.56

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -24.7	(c) -18.9	(d) -31.2	(e) 1.0	(f) -24.3	(g) -25.4	(h) 1.5	(i) -18.3	(j) 23.1	(k) 37.7	(l) 19.5	(m) 35.1	(n) 2.8	(o) 0	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 22.4	(c) 80.6	(d) 62.7	(e) 77.8	(f) 61.3	(g) 75.0	(h) 60.3	(i) 66.1	(j) 75.6	(k) 64.0	(l) 73.9	(m) 62.2	(n) 71.9	(o) 7.5	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 62.5	(b) 91.7	(c) 71.9	(d) 60.0	(e) 84.0	(f) 69.0	(g) 58.2	(h) 78.5	(i) 66.9	(j) 32.1	(k) 75.1	(l) 30.4	(m) 74.3	(n) 29.0	(o) 72.1	(p) 73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	16.1	13.5	-33.0	86.0	5.1	2.6	-36.6	87.9	-39.6	89.4	-42.5	90.9	-46.1	92.8
			-33.2	70.0	5.0	2.3	-36.8	71.7	-39.8	73.0	-42.6	74.3	-46.3	76.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.9	14.0	13.8	24.3	38.3	50.9	57.6	61.4	59.2	50.6	38.3	22.8	10.4
	DBStd	21.12	15.86	14.89	13.50	8.42	7.84	5.39	4.63	5.24	7.70	7.97	11.83	14.21
	HDD50	5877	1116	1013	797	359	85	7	0	3	84	369	815	1229
	HDD65	10280	1581	1433	1262	801	437	229	131	188	432	827	1265	1694
	CDD50	1106	0	0	0	8	113	234	353	289	102	7	0	0
	CDD65	34	0	0	0	0	1	5	18	9	1	0	0	0
	CDH74	801	0	0	0	6	102	130	320	162	81	0	0	0
	CDH80	105	0	0	0	0	15	12	53	14	11	0	0	0
(14) Wind	WSAvg	5.8	5.6	5.3	5.7	6.9	6.8	6.6	5.6	5.1	5.5	5.9	4.9	5.0
(15) Precipitation	PrecAvg	19.1	1.0	0.6	0.6	1.1	2.0	3.5	3.8	2.5	1.6	0.9	1.0	0.7
	PrecMax	27.8	3.9	1.4	1.5	2.4	3.6	6.1	8.3	4.0	4.4	2.4	3.0	1.7
	PrecMin	12.8	0.0	0.0	0.1	0.2	0.4	1.5	2.0	0.9	0.4	0.2	0.2	0.2
	PrecStd	3.3	0.9	0.4	0.4	0.6	0.9	1.4	1.7	0.8	0.9	0.6	0.9	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	48.7	56.2	73.3	82.2	81.6	85.1	81.8	81.6	68.4	51.1	45.6
		MCWB	38.6	39.2	43.7	52.2	55.9	61.1	65.8	65.4	62.5	53.0	42.9	36.7
	2%	DB	44.4	41.9	50.1	64.0	76.8	77.7	80.7	78.2	75.8	61.1	44.9	39.6
		MCWB	37.4	34.9	38.9	46.6	54.6	59.2	64.5	64.2	59.9	48.3	38.4	33.5
	5%	DB	39.9	38.2	45.4	57.9	71.9	74.3	77.4	75.3	70.6	56.3	40.7	34.8
		MCWB	34.4	32.8	36.8	43.6	52.3	57.8	63.3	62.7	57.2	45.6	35.4	30.2
	10%	DB	35.5	34.2	41.1	53.3	67.9	71.2	74.3	72.3	65.4	51.9	36.8	29.8
		MCWB	31.3	30.1	34.3	41.2	50.5	56.7	61.6	61.1	54.2	43.3	33.4	26.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	39.7	44.2	52.3	58.1	64.4	70.0	69.3	63.8	53.7	44.2	37.6
		MCDB	46.8	48.0	55.4	72.9	74.7	75.6	81.1	78.3	78.0	66.4	49.7	44.9
	2%	WB	37.5	35.6	40.4	47.4	55.9	61.8	66.8	66.2	60.9	49.6	39.2	33.7
		MCDB	44.2	40.8	48.9	61.7	71.6	73.4	76.5	75.1	73.9	58.9	43.5	39.1
	5%	WB	34.9	33.1	37.3	44.5	53.9	59.8	64.6	63.8	57.8	46.5	36.0	30.2
		MCDB	39.7	37.7	44.7	56.2	69.4	70.9	74.5	72.7	68.7	54.8	39.8	34.2
	10%	WB	31.5	30.3	34.6	41.9	51.8	58.1	62.7	62.0	55.4	43.8	33.6	26.8
		MCDB	35.5	34.2	40.5	52.1	65.4	68.2	72.4	70.6	63.8	50.6	36.5	29.6
(35) Mean Daily Temperature Range	MDBR		17.9	20.7	20.4	20.5	24.3	21.9	22.4	22.8	23.2	19.0	15.4	16.4
	5% DB	MCDBR	19.3	23.2	24.5	30.7	32.9	28.4	28.4	28.0	32.7	27.2	19.8	21.0
		MCWBR	13.8	17.1	15.3	16.5	15.0	13.2	15.2	15.6	17.9	16.1	14.1	16.3
	5% WB	MCDBR	19.3	22.4	23.1	28.8	28.8	24.2	26.7	25.9	30.7	25.4	18.1	20.6
		MCWBR	14.1	16.9	14.9	16.1	13.8	12.3	15.4	15.4	17.1	15.8	13.6	16.6
(40) Clear-Sky Solar Irradiance	taub		0.248	0.267	0.297	0.328	0.346	0.357	0.359	0.358	0.315	0.283	0.254	0.239
	taud		2.355	2.384	2.389	2.403	2.399	2.378	2.425	2.378	2.512	2.575	2.476	2.311
	Ebn at Noon		234	268	282	286	285	282	279	272	273	261	233	210
	Edh at Noon		16	23	29	32	35	36	34	33	25	19	15	14
(44) All-Sky Solar Radiation	RadAvg		270	585	1037	1456	1752	1838	1849	1543	1067	600	283	193
	RadStd		23	43	85	125	142	134	120	116	112	67	27	17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page